ADVANCED EXCEL · LESSON 1 OF 12

Getting Started with Excel for Data Analysis

Learn the Excel window, the words analysts use every day, how to lay out data so it can be analysed, the five data types Excel understands, and the shortcuts that save hours.

Live classes 1–2 | ✦ Excel for Microsoft 365, 2026 | Beginner | ⏱ 90 min

By the end of this lesson you can

- Name every part of the Excel window and explain what it does
- Use the core vocabulary (workbook, worksheet, cell, range, reference, formula, function) correctly
- Recognise structured data and fix a badly laid-out sheet
- Identify the five data types Excel stores, and spot text that only looks like a number or date
- Navigate and select data quickly with keyboard shortcuts

Datasets used

`employees_small.csv`

Why Excel still matters in 2026

Even with Python, SQL, Power BI and AI assistants, Excel is still the first tool almost every analyst opens. It is where business teams keep budgets, targets, trackers and exports from other systems. If you can clean, analyse and present data in Excel quickly, you can add value from day one in any data role.

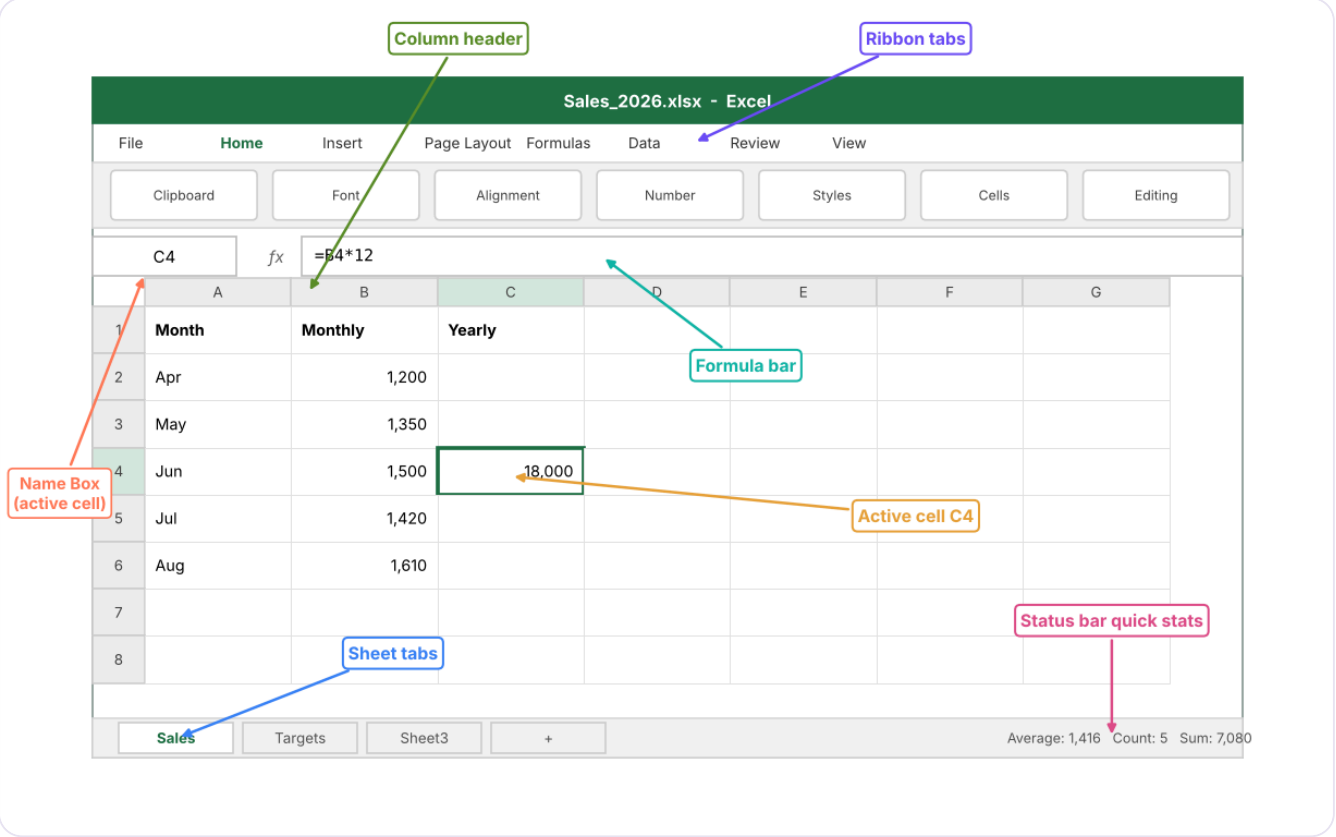
Excel is also the easiest place to *learn* data thinking. Every idea you meet here comes back later:

Idea in Excel	Same idea later in the course
A table with one header row	A SQL table, a pandas DataFrame, a Power BI table
SUMIFS, COUNTIFS	`GROUP BY` in SQL, `groupby()` in pandas, DAX measures
VLOOKUP / XLOOKUP	SQL `JOIN`, pandas `merge()`, Power BI relationships
Pivot table	SQL aggregation, `pivot_table()` in pandas, Power BI matrix
Conditional formatting	Highlighting in dashboards

LIVE CLASS **What was covered in class**

Live class 1 was the programme introduction. Live class 2 covered the basics of Excel, terminology, shortcuts, Excel properties, data types, data validation and formatting, and we started working on the first sheet. This lesson covers the first half; data validation and formatting continue in Lessons 2 and 3.

The Excel window, part by part



	A	B	C
1	Month	Monthly	Yearly
2	Apr	1,200	
3	May	1,350	
4	Jun	1,500	18,000
5	Jul	1,420	
6	Aug	1,610	
7			
8			

The main parts of the Excel window. The active cell is C4, and the formula bar shows the formula inside it.

Part	What it does	Tip
Ribbon	Commands grouped into tabs: Home, Insert, Formulas, Data, Review, View	Press `Alt` to see a key letter on every command
Name Box	Shows the address of the active cell or the name of a selected range	Type `D500` and press Enter to jump straight there
Formula bar	Shows what is really inside the cell: a value or a formula	The cell shows the *result*; the bar shows the *recipe*
Column headers	Letters A, B, C … XFD (16,384 columns)	Double-click the border between headers to auto-fit width
Row headers	Numbers 1 to 1,048,576	Click a row number to select the whole row
Sheet tabs	Each tab is one worksheet inside the workbook	Right-click a tab to rename, colour, move or copy it
Status bar	Quick sum, average and count of the selected cells	Right-click it to add Min, Max and Numerical Count

TIP

The status bar is a free calculator

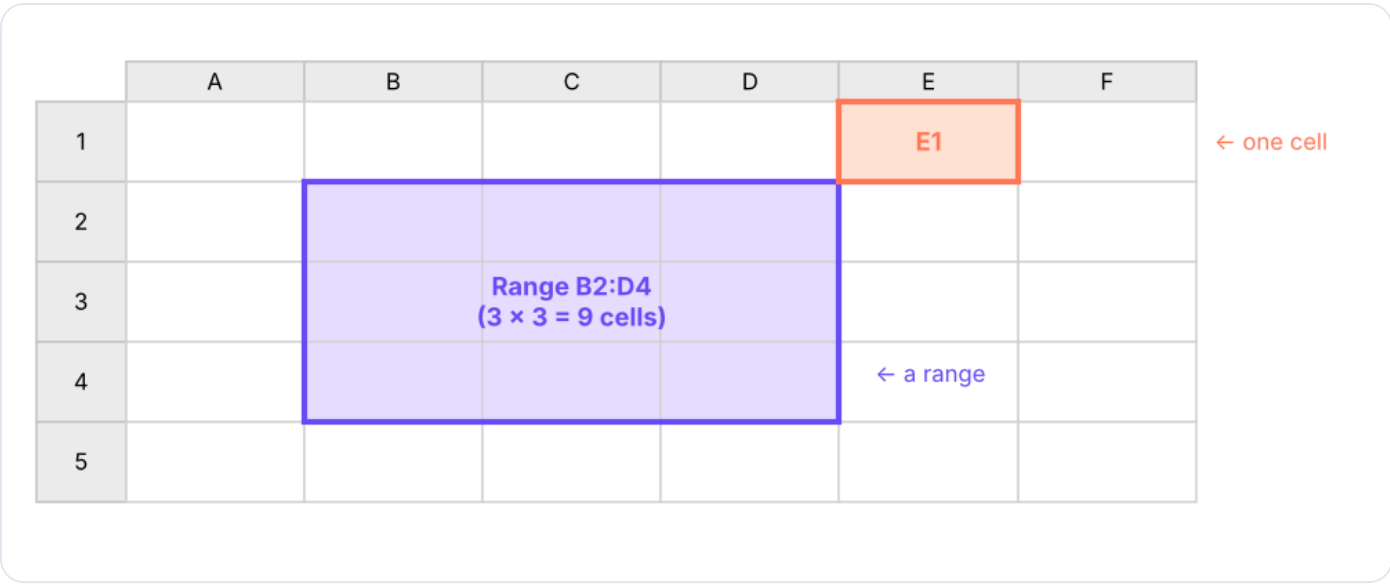
Select any numbers and look at the bottom-right corner. Excel shows the Average, Count and Sum without you writing a formula. It is the fastest way to sanity-check a total someone sends you.

The vocabulary every analyst uses

DEFINITION

Core terms

Workbook: the file (`.xlsx`). One workbook holds many worksheets. **Worksheet (sheet)**: one grid of rows and columns inside a workbook. **Cell**: one box where a row and a column meet, for example `C4` . **Cell reference (address)**: column letter + row number, such as `C4` . Used inside formulas to point at cells. **Range**: a rectangle of cells written as `first:last` , for example `A1:C10` (3 columns × 10 rows = 30 cells). **Formula**: anything that starts with `=` and calculates a result, such as `=B4*12` . **Function**: a built-in, named formula that takes inputs (arguments), such as `=SUM(B2:B6)` . **Argument**: an input you give a function, separated by commas: `=ROUND(B2, 1)` has two arguments.



How cells, ranges and references relate. A range is always written from its top-left cell to its bottom-right cell.

Excel limits worth knowing

Item	Limit in Excel 365
Rows per sheet	1,048,576
Columns per sheet	16,384 (A to XFD)
Characters in one cell	32,767
Characters in a formula	8,192
Arguments in one function	255
Decimal precision	15 significant digits

WATCH OUT **The 15-digit trap**

Excel keeps only 15 significant digits. If you type a 16-digit card number or a long ID, the last digits turn into zeros. Store long IDs, phone numbers, PIN codes and account numbers as **text** (format the column as Text first, or start the entry with an apostrophe `'`).

Structured data: the single most important habit

Excel will do almost anything, but it can only *analyse* data that is laid out as a clean table. Analysts call this **structured** or **tidy** data.

STEP BY STEP

The six rules of a structured table

1. **One header row** at the top, with a unique, short name for every column.
2. **One record per row**, for example one order, one employee or one day.
3. **One kind of information per column**: dates only in the date column, numbers only in the amount column.
4. **No blank rows or blank columns** inside the table.
5. **No merged cells** and no totals typed in the middle of the data.
6. **Raw values, not decorations**: write `Delhi`, not `Delhi ✔ (checked)`.

Here is a typical "report-style" sheet that looks fine to a human but is hard to analyse:

Before: a report, not a table

	A	B	C	D
1	**Sales Report – Q1**			
2	*Region: North*			
3	Name	Jan	Feb	Mar
4	Asha Rao	12,000	15,500	n/a
5	Ravi Verma	9,800		11,250
6				
7	*Region: South*			
8	Name	Jan	Feb	Mar
9	Meera Iyer	14,250	13,900	16,100
10	**TOTAL**	36,050	29,400	27,350

Problems: a title row above the headers, the region written as a heading instead of a column, a blank row, the header row repeated, numbers stored as text with commas, n/a mixed into a number column, and a total row inside the data.

The same information as a structured table:

After: a structured table (long format)

	A	B	C	D
1	**Region**	**Name**	**Month**	**Sales**
2	North	Asha Rao	Jan	12000
3	North	Asha Rao	Feb	15500
4	North	Ravi Verma	Jan	9800
5	North	Ravi Verma	Mar	11250
6	South	Meera Iyer	Jan	14250
7	South	Meera Iyer	Feb	13900
8	South	Meera Iyer	Mar	16100
9	**Total**			92800

```
D9   =SUM(D2:D8)
```

Now a pivot table, a SUMIFS, a chart or Power BI can use the data immediately. The total sits **below** the table, not inside it, and is calculated, not typed.

> **TIP** **Long format beats wide format**
>
> In the "before" sheet each month is a separate column (wide format). In the "after" sheet the month is a value in one `Month` column (long format). Long format is what pivot tables, SQL and Power BI expect. In Lesson 12 you will see how Power Query's **Unpivot** turns wide data into long data in one click.

The five data types Excel understands

Every cell holds exactly one type of value. Knowing the type explains most "my formula doesn't work" problems.

Type	Examples	Default alignment	Notes
Number	`42`, `3.14`, `-500`, `18%`	Right	Percentages and currency are numbers with a format
Text (string)	`Delhi`, `E101`, `'00451`	Left	Anything Excel cannot read as a number
Date / time	`27-04-2026`, `14:30`	Right	Stored as a serial number: days since 1 Jan 1900
Logical (Boolean)	`TRUE`, `FALSE`	Centre	Result of comparisons like `=B2>100`
Error	`#N/A`, `#DIV/0!`, `#VALUE!`	Centre	Tells you a formula has a problem

Excel's own functions can tell you the type of a value. Each formula in column B below checks the value in column A:

Checking data types with functions

	A	B	C	D
1	Value	ISNUMBER	ISTEXT	TYPE (1=number, 2=text, 4=logical)
2	42	TRUE	FALSE	1
3	Delhi	FALSE	TRUE	2
4	00451	FALSE	TRUE	2
5	27-04-2026	TRUE	FALSE	1
6	TRUE	FALSE	FALSE	4
7	1500	FALSE	TRUE	2

B2 `=ISNUMBER(A2)`

C2 `=ISTEXT(A2)`

D2 `=TYPE(A2)`

Each pattern is copied down/across to the other cells in its column.

Notice two important results:

- The **date** in row 5 is a number. Excel stores 27-04-2026 as the serial number 46139, which is why you can subtract one date from another to get days.
- `'1500` in row 7 is **text**, even though it looks like a number. `SUM` ignores it. Imported data often has "numbers stored as text"; Excel flags them with a small green triangle.

Dates are numbers underneath

	A	B	C
1	Date	Same cell shown as a number	Days until 31-Dec-2026
2	27-04-2026	46139	248
3	01-01-2026	46023	364

```
B2   =A2
C2   =DATE(2026,12,31)-A2
B3   =A3
C3   =DATE(2026,12,31)-A3
```

COMMON MISTAKES **Fixing numbers stored as text**

- Select the cells, click the yellow warning icon, choose **Convert to Number**.
- Or use **Data → Text to Columns → Finish** on the column.
- Or in a formula: `=VALUE(A7)` or `=A7*1`.
- Remove hidden spaces first with `=TRIM(A7)` (and `=CLEAN()` for line breaks from web exports).

Error values and what they mean

Error	Meaning	Typical cause
`#DIV/0!`	Division by zero	`=A2/B2` where B2 is empty or 0
`#N/A`	Value not available	A lookup could not find the value
`#VALUE!`	Wrong type of argument	Doing maths with text, such as `="abc"*2`
`#REF!`	Invalid reference	You deleted a row or column a formula used
`#NAME?`	Excel does not recognise a name	A typo in a function name, or text without quotes
`#NUM!`	Invalid number	`=SQRT(-1)`
`#SPILL!`	A dynamic array has no room	Cells below the formula are not empty (Lesson 6)
`#####`	Not an error	The column is too narrow to show the value

Excel properties: what makes a workbook "smart"

These behaviours make Excel more than a calculator:

1. **Automatic recalculation**: change an input and every dependent formula updates instantly (Formulas → Calculation Options can switch this to Manual for huge files; press F9 to

recalculate).

2. **Relative references**: copy `=B2*12` down and it becomes `=B3*12`, `=B4*12` automatically (Lesson 4 covers $ to lock references).
3. **AutoFill**: drag the small square (fill handle) at the bottom-right of a selection to continue patterns: `Jan → Feb → Mar`, `1, 2, 3`, `Mon, Tue`, dates and formulas.
4. **Flash Fill** (`Ctrl + E`): type one or two examples of a pattern (such as first names taken from full names) and Excel fills the rest.
5. **Excel Tables** (`Ctrl + T`): turn a range into a named table that expands automatically, keeps formatting, and allows readable formulas like `=SUM(Sales[Amount])`.
6. **File properties**: File → Info shows the author, size, last-modified date and lets you inspect the workbook for hidden data before sharing.

2026 UPDATE **What's new in Excel (2025–2026)**

- **Copilot in Excel** can explain formulas, suggest formula columns, highlight insights and create charts from a plain-English request (Microsoft 365 Copilot licence required).
- **Python in Excel** lets you type `=PY(` and run pandas code inside a cell; the code runs in Microsoft's cloud.
- **Dynamic array functions** such as `UNIQUE`, `SORT`, `FILTER`, `XLOOKUP`, `TEXTSPLIT`, `GROUPBY` and `PIVOTBY` are now standard in Microsoft 365.
- **Checkbox cells** (Insert → Checkbox) store TRUE/FALSE and are useful for trackers. You will use each of these later in the Excel track.

Keyboard shortcuts that save hours

You do not need to memorise 100 shortcuts. Learn these 25 and you will already be faster than most office users. (Mac users: replace `Ctrl` with `Cmd` for most of them.)

Task	Shortcut
Jump to the edge of the data	`Ctrl + Arrow key`
Select to the edge of the data	`Ctrl + Shift + Arrow key`
Select the whole current table	`Ctrl + A` (inside the data)
Go to cell A1 / last used cell	`Ctrl + Home` / `Ctrl + End`
Select a whole column / row	`Ctrl + Space` / `Shift + Space`
Insert / delete rows or columns	`Ctrl + Shift + +` / `Ctrl + -`
Edit the active cell	`F2`
Lock a reference (A1 → A$1 → $A1)	`F4` while editing a formula
Fill down / fill right	`Ctrl + D` / `Ctrl + R`
Enter the same value in all selected cells	type, then `Ctrl + Enter`
AutoSum	`Alt + =`
Today's date / current time	`Ctrl + ;` / `Ctrl + Shift + ;`
Create an Excel Table	`Ctrl + T`
Toggle filters	`Ctrl + Shift + L`
Format Cells dialog	`Ctrl + 1`
Show formulas instead of results	`Ctrl + ` `
Flash Fill	`Ctrl + E`
Paste Special	`Ctrl + Alt + V`
Find / Replace	`Ctrl + F` / `Ctrl + H`
New sheet	`Shift + F11`
Move between sheets	`Ctrl + Page Up / Page Down`
Hide rows / columns	`Ctrl + 9` / `Ctrl + 0`
Undo / Redo	`Ctrl + Z` / `Ctrl + Y`
Repeat last action	`F4` (outside a formula)
Show key tips on the ribbon	`Alt`

TIP **Practise with real data**

Open `employees_small.csv` (download it from the top of this lesson). Click any cell inside the data, press `Ctrl + A`, then `Ctrl + T` to make it a Table. Press `Ctrl + Down` to jump to the last row and `Ctrl + Home` to come back. Select the salary column and read the Sum and Average in the status bar.

Your first worksheet, step by step

Let's build a small salary sheet with the 20-row employee file. The first rows look like this:

Python

```
import pandas as pd
emp = pd.read_csv("employees_small.csv")
emp.head(6)
```

OUTPUT

	emp_id	name	department	city	join_date	salary	rating
0	E101	Kavya Iyer	HR	Mumbai	2016-02-28	97000	4
1	E102	Imran Reddy	Finance	Delhi	2019-12-14	157000	3
2	E103	Harpreet Kulkarni	Finance	Mumbai	2024-10-05	131000	4
3	E104	Diya Iyer	HR	Bengaluru	2024-01-02	90000	4
4	E105	Simran Iyer	HR	Bengaluru	2017-08-03	139000	4
5	E106	Saanvi Rao	HR	Delhi	2025-02-18	83000	4

6 rows × 7 columns

STEP BY STEP **Build it**

1. **Open** the CSV in Excel (File → Open, or Data → From Text/CSV).
2. Press `Ctrl + T` to convert it to a Table, and name it `Emp` (Table Design → Table Name).
3. In a new column `annual_salary`, type = then click the salary cell and type `*12`. Because it is a Table, Excel fills the whole column with `=[@salary]*12`.
4. Below the data, add summary formulas: total salary, average salary, number of employees.
5. Format the salary columns as `#,##0` with `Ctrl + 1` (Lesson 3 explains every format).
6. Freeze the header row: View → Freeze Panes → Freeze Top Row.

First worksheet: summary formulas (data on sheet Emp)

	A	B	C
1	**Measure**	**Formula result**	**What the formula does**
2	Employees	20	Counts non-empty IDs
3	Total monthly salary	1,978,000	Adds the salary column
4	Average monthly salary	98,900	Total ÷ count
5	Highest salary	157,000	Largest value
6	Lowest salary	45,000	Smallest value

```
B2   =COUNTA(Emp!A2:A21)
B3   =SUM(Emp!F2:F21)
B4   =AVERAGE(Emp!F2:F21)
B5   =MAX(Emp!F2:F21)
B6   =MIN(Emp!F2:F21)
```

The results above were calculated from the real file. Check them yourself in Python:

Python

```python
s = emp["salary"]
print("Employees:", s.count())
print("Total:", f"{s.sum():,}", "| Average:", f"{s.mean():,.0f}", "| Max:", f"{s.max():,}",
"| Min:", f"{s.min():,}")
```

OUTPUT

```
Employees: 20
Total: 1,978,000 | Average: 98,900 | Max: 157,000 | Min: 45,000
```

Saving and sharing the right way

Format	Use it when
`.xlsx`	Normal workbooks (default)
`.xlsm`	Workbooks that contain macros (VBA)
`.xlsb`	Very large workbooks; binary format opens and saves faster
`.csv`	Plain data to move between systems (Python, SQL, Power BI). Keeps only one sheet and no formatting or formulas
`.pdf`	Read-only reports for people who should not edit

WATCH OUT **CSV and leading zeros**

When you open a CSV directly, Excel guesses types and may drop leading zeros (`00451` → `451`) or turn codes like `1-2` into dates. Use **Data → From Text/CSV** and set the column type to Text in the preview to keep values exactly as they are.

Summary

KEY TAKEAWAYS

- The **cell** shows the result; the **formula bar** shows what is really inside.
- A **range** is written top-left to bottom-right, for example `B2:D4`.
- Analyse only **structured data**: one header row, one record per row, one type per column, no blanks, no merged cells, no totals inside.
- Excel has five value types: **number, text, date/time (a number underneath), logical, error**.
- Long IDs and codes must be stored as **text** to avoid the 15-digit and leading-zero traps.
- Learn the navigation shortcuts first: `Ctrl + Arrow`, `Ctrl + Shift + Arrow`, `Ctrl + T`, `Ctrl + 1`, `F2`, `F4`.

Quick quiz

6 questions · tap an answer to check it

1. A cell shows `18,000`, but the formula bar shows `=B4*12`. What is inside the cell?

- **A** The number 18,000
- **B** A formula whose result is 18,000
- **C** Text that looks like a number
- **D** An error

Answer: B. The cell displays the result; the formula bar shows what is stored. This cell stores a formula.

2. How many cells are in the range `B2:D5`?

- **A** 8
- **B** 9
- **C** 12
- **D** 15

Answer: C. Columns B, C, D = 3 columns; rows 2 to 5 = 4 rows; 3 × 4 = 12 cells.

3. Which of these breaks the rules of structured data?

- **A** One header row with unique names
- **B** A merged "Region" heading above each group of rows
- **C** One record per row
- **D** A date column that contains only dates

Answer: B. Region should be its own column, with the value repeated on every row. Merged headings make the data impossible to sort, filter or pivot.

4. Why does `SUM` ignore some numbers imported from a CSV file?

A SUM only works on Excel Tables

B The values are stored as text

C The numbers have too many decimals

D The column is too narrow

Answer: B. Numbers stored as text are skipped by SUM. Convert them with Text to Columns, VALUE() or Convert to Number.

5. What does Excel store when you type the date 27-04-2026?

A The text "27-04-2026"

B A serial number counting days from 1 January 1900

C Three separate numbers for day, month and year

D A logical value

Answer: B. Dates are numbers formatted to look like dates. That is why date arithmetic works.

6. Which shortcut turns a range into an Excel Table?

A Ctrl + E

B Ctrl + 1

C Ctrl + T

D Ctrl + Shift + L

Answer: C. Ctrl + T creates a Table. Ctrl + E is Flash Fill, Ctrl + 1 opens Format Cells, Ctrl + Shift + L toggles filters.

EXERCISE **Clean a messy sheet**

Look at the "Before" report in this lesson. Write down, in order, the steps you would take in Excel to turn it into the "After" table.

Show solution (worked answer)

1. Delete the title row and the blank row.
2. Insert a new first column called `Region` and fill it with North or South for each person; delete the "Region:" heading rows and the repeated header row.
3. Delete the TOTAL row (totals belong below or in a pivot table).
4. Replace `n/a` with an empty cell (Ctrl + H, replace `n/a` with nothing).
5. Convert the Jan–Mar columns to real numbers (Text to Columns or Convert to Number).
6. Unpivot the month columns into `Month` and `Sales` (Power Query → Unpivot Other Columns, Lesson 12) so each row is one person-month.
7. Press Ctrl + T to make it a Table and add the total with `=SUM()` below it.

EXERCISE **Spot the data type**

For each entry, say which data type Excel stores: `₹1,250`, `'2026`, `15:45`, `=5>3`, `#N/A`, `Pune`.

Show solution (worked answer)

`₹1,250` is a number (currency format). `'2026` is text because of the apostrophe. `15:45` is a time, which is a number (0.65625 of a day). `=5>3` returns the logical value TRUE. `#N/A` is an error. `Pune` is text.

INTERVIEW CORNER

Questions you may be asked

Q: What is the difference between a formula and a function? A formula is any expression that starts with = (for example `=A2*B2`). A function is a named, built-in calculation used inside formulas (for example `SUM`, `VLOOKUP`). Every function is used in a formula, but not every formula uses a function.

Q: How does Excel store dates, and why does it matter? As serial numbers (days since 1-Jan-1900; times are fractions of a day). It means you can subtract dates to get durations, add days to a date, and sort dates correctly. If a date is stored as text, none of that works.

Q: What makes data "analysis-ready"? One header row, one record per row, one data type per column, no blank rows or merged cells, no subtotals mixed into the data, and consistent spelling of categories.

Practise more online

Free, trusted places to test yourself on this topic. Links open the original site in a new tab.